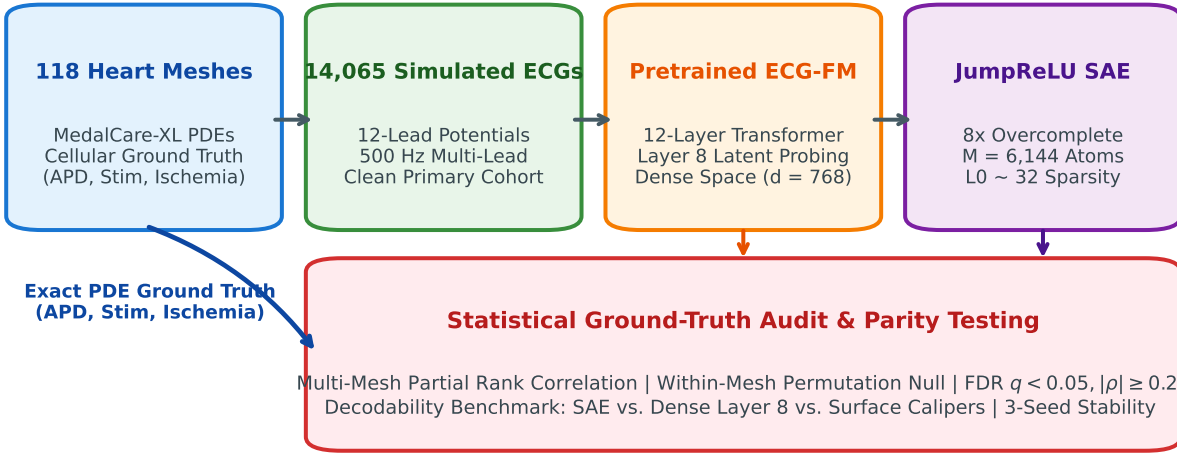
118 Heart Meshes
MedalCare-XL PDEs
Cellular Ground Truth
(APD, Stim, Ischemia)
14,065 Simulated ECGs
12-Lead Potentials
500 Hz Multi-Lead
Clean Primary Cohort
Pretrained ECG-FM
12-Layer Transformer
Layer 8 Latent Probing
Dense Space (d = 768)
JumpReLU SAE
8x Overcomplete
M = 6,144 Atoms
L0 ~ 32 Sparsity
Exact PDE Ground Truth
(APD, Stim, Ischemia)
Statistical Ground-Truth Audit & Parity Testing
Multi-Mesh Partial Rank Correlation | Within-Mesh Permutation Null | FDR $q < 0.05$, $|\rho| \geq 0.25$
Decodability Benchmark: SAE vs. Dense Layer 8 vs. Surface Calipers | 3-Seed Stability